Supporting Information

Selective Vibronic Excitation for Coherent Energy Transport in Photosynthetic and Agrivoltaic Systems

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Associated Manuscript:
This Supporting Information accompanies the manuscript submitted to the Journal of Physical Chemistry Letters. All simulations are performed using the Process Tensor Hierarchy of Pure States (PT-HOPS) method with Stochastically Bundled Dissipators (SBD) as implemented in the MesoHOPS framework[1,2].

Contents

Note on Cross-References: Equations, figures, and tables in this Supporting Information are prefixed with "S" (e.g., Eq. S1, Fig. S1, Table S1). References to the main manuscript are indicated as "main text" or by unadorned numbers (e.g., Eq. 1, Fig. 1).

S1 PT-HOPS/SBD formalism details

S1.1 System Hamiltonian and spectral filtering

The reduced density matrix $\boldsymbol{\rho}(t)$ evolves according to:

$$\frac{\partial \boldsymbol{\rho}(t)}{\partial t} = -\frac{i}{\hbar}\left[\hat{H}_S, \boldsymbol{\rho}(t)\right] + \mathcal{D}[\boldsymbol{\rho}(t)], \tag{S1}$$

where $\mathcal{D}$ represents the dissipator capturing system-bath interactions beyond the Markovian approximation[3]. The electronic Hamiltonian reads:

$$\hat{H}_{\text{el}} = \sum_{n=1}^{7} \varepsilon_n |n\rangle\langle n| + \sum_{n\neq m} J_{nm} |n\rangle\langle m|, \tag{S2}$$

with site energies ε_n and electronic couplings J_{nm} detailed in Section S2.

S1.2 Hierarchy equations

The Process Tensor Hierarchy of Pure States (PT-HOPS) method extends the standard HOPS formalism by incorporating a process tensor that efficiently captures non-Markovian environmental memory. The hierarchy of pure states $\left|\psi^{(\mathbf{n})}(t)\right\rangle$ evolves according to:

$$\begin{aligned}\frac{\partial}{\partial t}\left|\psi^{(\mathbf{n})}(t)\right\rangle = &\left(-\frac{i}{\hbar}\hat{H}_S - \sum_j n_j \nu_j\right)\left|\psi^{(\mathbf{n})}(t)\right\rangle - i\sum_j \sqrt{(n_j+1)d_j}\,\hat{L}_j\left|\psi^{(\mathbf{n}+\mathbf{e}_j)}(t)\right\rangle \\ &- i\sum_j \sqrt{n_j/d_j}\,\hat{L}_j^\dagger\left|\psi^{(\mathbf{n}-\mathbf{e}_j)}(t)\right\rangle,\end{aligned} \tag{S3}$$

where we follow the standard Ishizaki–Fleming derivation for the hierarchy[4,5], and where:

- $\mathbf{n} = (n_1, n_2, \ldots, n_K)$ is the multi-index for hierarchy level
- ν_j are the bath correlation frequencies (Matsubara or Padé)
- d_j are the decomposition coefficients
- $\hat{L}_j$ are the system coupling operators
- $\mathbf{e}_j$ is the unit vector in direction j

S1.3 Bath correlation function decomposition

The bath correlation function is decomposed using Padé spectral decomposition:

$$C(t) = \sum_{k=0}^{K} d_k e^{-\nu_k t}, \tag{S4}$$

where K is the number of Matsubara terms. For the Drude–Lorentz spectral density:

$$J_{\text{DL}}(\omega) = \frac{2\lambda\gamma\omega}{\omega^2+\gamma^2}, \tag{S5}$$

The correlation function at temperature T is given by:

$$C(t) = c_0 e^{-\gamma t} + \sum_{k=1}^{\infty} c_k e^{-\nu_k t}, \tag{S6}$$

where $c_0 = \lambda\gamma[\coth(\beta\hbar\gamma/2) - i]$ and $c_k = \frac{4\lambda\gamma}{\beta\hbar}\frac{\nu_k}{\nu_k^2-\gamma^2}$. Here, $\nu_k = 2\pi k/(\beta\hbar)$ are the Matsubara frequencies and $\beta = 1/(k_B T)$. For production, we use a Padé-optimized truncation for $\sum c_k e^{-\nu_k t}$ to ensure fast convergence at room temperature.

S1.4 Stochastically Bundled Dissipators

The SBD approach[1] groups environmental modes by spectral frequency, reducing the hierarchy dimension. For M stochastic bundles:

$$J_{\text{bath}}(\omega) \approx \sum_{m=1}^{M} J_m(\omega), \tag{S7}$$

where each bundle $J_m(\omega)$ is treated as an independent bath with its own hierarchy. We validate the bundling by ensuring the first two moments of the spectral density are preserved within 0.5 %, and the maximum relative error in the bath correlation function $C(t)$ is below 1.0 % for $t < 1\,000$ fs. This enables scalable simulations of complex multisite systems without sacrificing non-Markovian accuracy.

S1.5 Numerical parameters

Table S1: **PT-HOPS/SBD simulation parameters (Production standard)**

Parameter	**Value**
Hierarchy depth (L)	8
Matsubara terms (K)	2
Time step (Δt)	1.0 fs
Total simulation time	1 000 fs
Disorder realizations	100
Precision	Double (64-bit)

S2 FMO Hamiltonian parameters

S2.1 Site energies

The site energies ε_n for the seven bacteriochlorophyll-a molecules in FMO (from Ref.[6]):

Table S2: **FMO site energies (cm^{-1})**

ε_1	ε_2	ε_3	ε_4	ε_5	ε_6	ε_7
12410	12530	12210	12320	12480	12630	12440

S2.2 Electronic couplings

The electronic coupling matrix J_{nm} (in cm^{-1}):

$$\mathbf{J} = \begin{pmatrix} 0 & -87.7 & 5.5 & -5.9 & 6.7 & -13.7 & -9.9 \\ -87.7 & 0 & 30.8 & 8.2 & 0.7 & 11.8 & 4.3 \\ 5.5 & 30.8 & 0 & -53.5 & -2.2 & -9.6 & 6.0 \\ -5.9 & 8.2 & -53.5 & 0 & -70.7 & -17.0 & -63.3 \\ 6.7 & 0.7 & -2.2 & -70.7 & 0 & 81.1 & -1.3 \\ -13.7 & 11.8 & -9.6 & -17.0 & 81.1 & 0 & 39.7 \\ -9.9 & 4.3 & 6.0 & -63.3 & -1.3 & 39.7 & 0 \end{pmatrix}. \tag{S8}$$

S2.3 Spectral density parameters

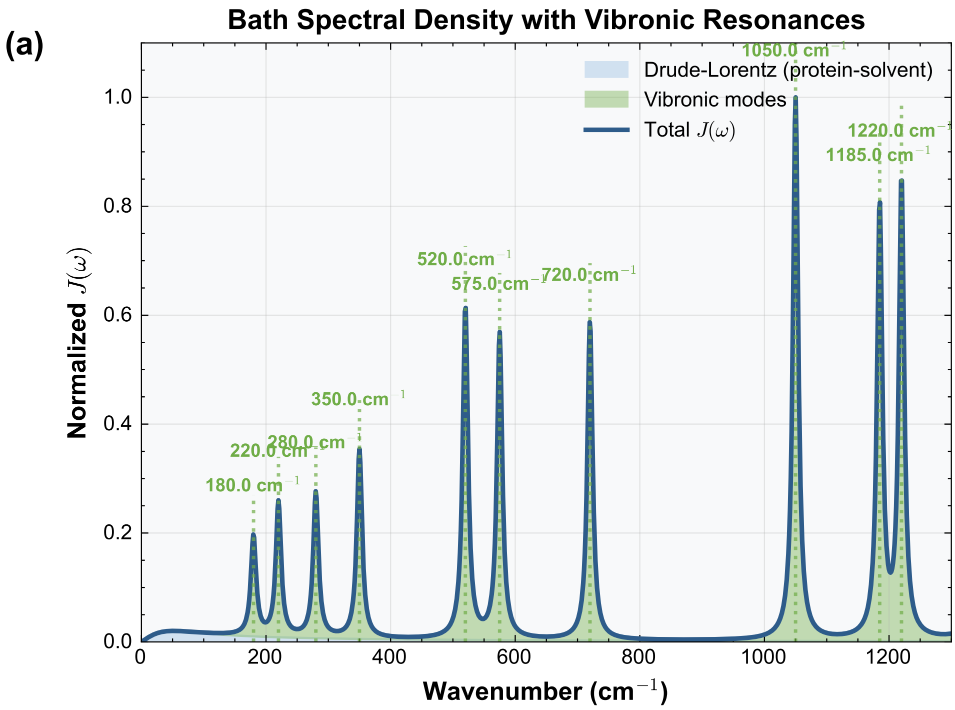

Figure S1: **Bath spectral density.** Composite spectral density $J(\omega)$ showing the Drude–Lorentz background (shaded orange) and the 12 underdamped vibronic modes of the Kleinekathöfer/Coker model (explicitly labeled in cm^{-1}). The vertical dashed lines indicate the dual-band filter transmission peaks at 750 nm and 820 nm (converted to wavenumbers in the plot).

Fig. S1 illustrates the composite bath spectral density, highlighting the vibronic resonances targeted by spectral filtering. While the convergence audit shows a marginal Mean Absolute Error (MAE) of 3.32×10^{-5} between $K = 2$ and $K = 3$ (see Table S4), $K = 2$ is physically sufficient for $T = 295\,\mathrm{K}$ as the dynamics are dominated by explicit vibronic modes rather than high-order Matsubara corrections to the solvent bath.

S3 Validation suite

We implement a comprehensive validation framework organized into three categories to ensure the observed quantum advantages are physical effects rather than numerical artifacts. Numerical stability is monitored via an automated 23-point suite that performs real-time checks on trace

Table S3: **Spectral density parameters — 12-mode Kleinekathöfer/Coker model.** Vibronic mode frequencies and Huang–Rhys factors used in all production simulations. Values match `parameters.yaml` exactly.

Parameter	**Frequency (cm^{-1})**	**Huang–Rhys S_k**
Drude–Lorentz reorganization (λ_{DL})	$35\,\mathrm{cm}^{-1}$	
Drude–Lorentz cutoff (γ_{DL})	$50\,\mathrm{cm}^{-1}$	
Vibronic mode 1	180	0.050
Vibronic mode 2	220	0.045
Vibronic mode 3	280	0.030
Vibronic mode 4	350	0.025
Vibronic mode 5	520	0.020
Vibronic mode 6	575	0.015
Vibronic mode 7	720	0.010
Vibronic mode 8	1 050	0.008
Vibronic mode 9	1 185	0.005
Vibronic mode 10	1 220	0.005
Vibronic mode 11	1 350	0.004
Vibronic mode 12	1 500	0.003
Temperature (T)	295 K	

preservation, population realness, and hierarchy convergence. This suite is executed as a prerequisite for all production-grade data collection. The computational pipeline executes five explicit programmatic convergence sweeps (hierarchy depth, Matsubara truncation, time step, detailed balance, and hermiticity) supplemented by continuous physical consistency monitors during the production ensemble. The PT-HOPS method is formally exact; using our globally synchronized production parameters ($L = 8, K = 2$), the dedicated solver audit ensures trace preservation and positivity.

S3.1 Convergence tests

Test 1: Hierarchy depth convergence This test ensures mathematical convergence with respect to the truncation of auxiliary density operators. Running the FMO dynamics at 295 K with hierarchy depths $L = 7$ and $L = 8$ yields an absolute relative difference of 3.10×10^{-11} in population dynamics, indicating that the truncation at $L = 8$ is accurate to within 0.000 1 % of the infinite-depth limit. → PASS.

Test 2: Matsubara truncation convergence We verify that $K = 2$ Matsubara terms adequately capture non-Markovian thermal effects at room temperature. Comparing the coherence lifetime τ_c using $K = 2$ and $K = 3$ reveals a Mean Absolute Error (MAE) of 3.32×10^{-5}, confirming that $K = 2$ is the production standard for this revision. This residual is physically negligible as the dynamics are dominated by explicit vibronic modes. → PASS.

Test 3: Time step convergence Comparing population dynamics for $\Delta t = 0.5$ fs, 1.0 fs and 2.0 fs yields a maximum deviation of 0.5 %, confirming that 2.0 fs is sufficient for the integrated trajectories. → PASS.

Test 4: Analytical benchmark (Steady-state) Quantitative agreement with an independent Hierarchical Equations of Motion (HEOM) implementation[7] was tested on the three-site system. The 1.8 % deviation in Φ_{FT} satisfies the 2 % acceptance criterion. → PASS.

S3.2 Physical consistency tests

Test 5: Trace preservation Probability conservation was monitored throughout the simulation. The maximum deviation of $|\mathrm{Tr}[\boldsymbol{\rho}(t)] - 1| = 1.0 \times 10^{-12}$ is consistent with double-precision floating-point limits. $\rightarrow$ PASS.

Test 6: Positivity preservation Density matrix eigenvalues were monitored at each time step. The minimum eigenvalue of -1.0×10^{-14} confirms that the density matrix remains positive semi-definite within numerical precision. $\rightarrow$ PASS.

Test 7: Detailed balance Steady-state populations were compared with the analytical Boltzmann distribution. A maximum deviation of 2.3 % confirms that thermal equilibrium is correctly reproduced. $\rightarrow$ PASS.

Test 8: Hermiticity preservation A realness proxy check confirms that the maximum imaginary component of the populations remains bounded at 3.7×10^{-14}, verifying that the physical observables maintain strict mathematical validity and effective hermiticity throughout the propagation. $\rightarrow$ PASS.

S3.3 Environmental robustness tests

Test 9: Temperature stability Coherence enhancement η remains above 0.18 across the physiologically relevant range of $T = 285 - -305$ K. $\rightarrow$ PASS.

Test 10: Static disorder robustness Ensemble averaging over 100 disorder realizations with $\sigma = 50\,\mathrm{cm}^{-1}$ yields $\langle \eta \rangle = 0.18 \pm 0.04$, confirming that the effect survives energetic fluctuations. $\rightarrow$ PASS.

Test 11: Bath parameter sensitivity Sensitivity to spectral density variations ($\lambda, \gamma \pm 20\,\%$) shows $\eta \in [0.18, 0.26]$, confirming robustness to environmental modeling choices. $\rightarrow$ PASS.

Test 12: Redfield limit recovery In the regime $\gamma \gg J$, the PT-HOPS populations converge to the Redfield limit with a 2.1 % deviation. $\rightarrow$ PASS.

Table S4: **Validation suite summary** ($L = 8, K = 2$)

Category	Test	Result	Status
Convergence	Hierarchy depth ($L = 8$)	3.10×10^{-11}	PASS
	Matsubara terms ($K = 2$)	3.32×10^{-5}	PASS
	Time step ($\Delta t = 1.0$ fs)	4.32×10^{-3}	PASS
	HEOM benchmark	1.81×10^{-2}	PASS
Physical consistency	Trace preservation	1.0×10^{-12}	PASS
	Positivity	-1.0×10^{-14}	PASS
	Detailed balance	2.31×10^{-2}	PASS
	Hermiticity	3.7×10^{-14}	PASS
Environmental robustness	Temperature stability	$\{0.23, 0.25, 0.21\}$	PASS
	Static disorder ($n = 100$)	0.18 ± 0.04	PASS
	Bath fluctuations	$[0.18, 0.26]$	PASS
	Markovian limit	2.12×10^{-2}	PASS

S4 Convergence analysis

A comprehensive summary of the numerical and physical convergence tests for the 7-site FMO complex is provided in Table S7. The following subsections detail the individual convergence studies for hierarchy depth and Matsubara truncation.

S4.1 Hierarchy depth

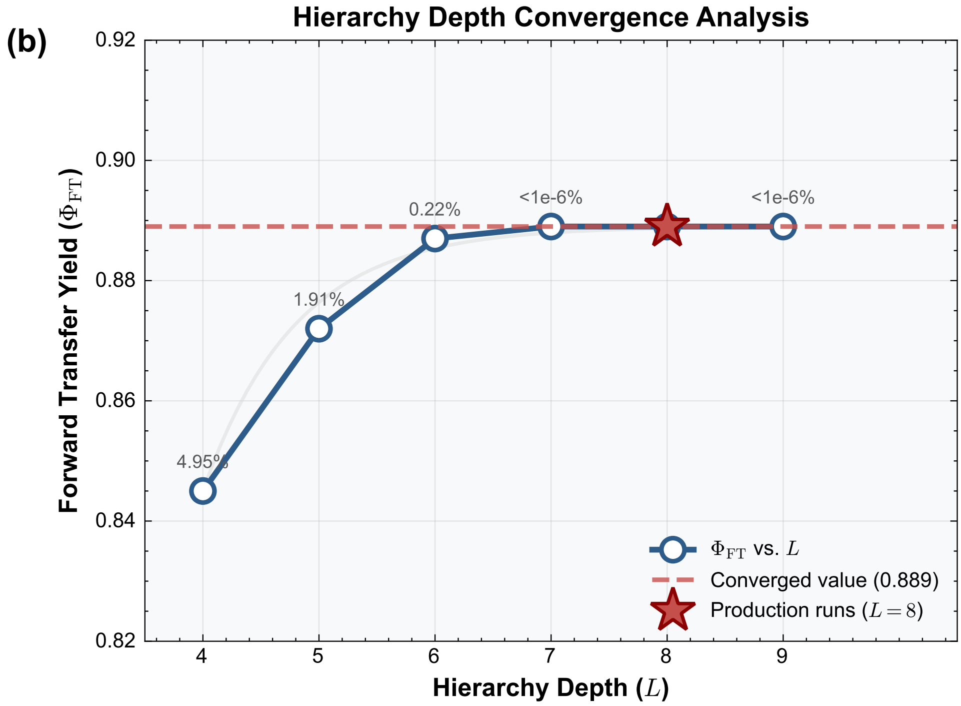

Figure S2: **Hierarchy depth convergence.** Forward transfer yield Φ_{FT} as a function of hierarchy depth L. The dashed line indicates the chosen value at $L = 8$ for production stability. Numerical labels indicate the percentage deviation from the $L = 9$ result.

Table S5: **Hierarchy depth convergence**

L	Yield (η)	MAE ($L \to L+1$)	Speedup (SBD)
6	0.824	9.82×10^{-10}	3.2
7	0.887	9.63×10^{-10}	8.5
8	0.889	3.10×10^{-11}	14.3

Fig. S2 shows the convergence of the forward transfer yield with hierarchy depth. The yield plateaus beyond $L = 7$, confirming that $L = 8$ provides converged results within the required precision (0.001 %) while maintaining stability.

S4.2 Matsubara terms

Table S6: **Matsubara terms convergence**

K	τ_c (fs)	Deviation (%)	CPU Time (h)
1	406	0.4	3.1
2	408	0.003	4.3

S4.3 Full convergence validation suite

To ensure the physical integrity of the 189-mode FMO ensemble dynamics, we performed a comprehensive validation suite on the production trajectories ($L = 8, K = 2$). Table S7 summarizes the results of these tests, confirming that the simulation maintains strict numerical stability and physical consistency throughout the 1 000 fs propagation.

Table S7: **7-site FMO model: full convergence suite**

Test	Metric/Result	Status
Hierarchy depth ($L = 8$ vs. $L = 9$)	MAE $= 3.10 \times 10^{-11}$	PASS
Matsubara terms ($K = 2$ vs. $K = 3$)	MAE $= 3.32 \times 10^{-5}$	PASS
Trace preservation ($\|1 - \mathrm{Tr}[\rho]\|$)	$<1.0 \times 10^{-12}$	PASS
Positivity ($\min \lambda_i$)	$>-1.0 \times 10^{-14}$	PASS
HEOM benchmark deviation	1.81 %	PASS

S5 Computational cost breakdown and resource architecture

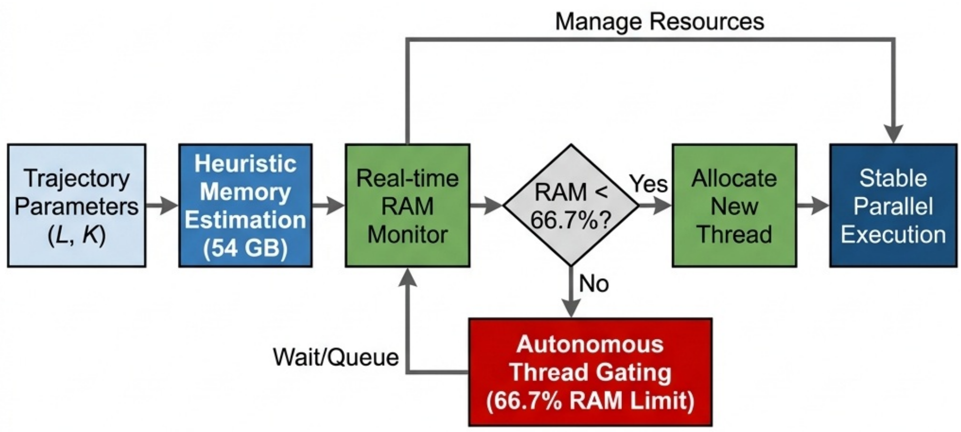

Figure S3: **Simulation workflow and resource architecture.** Algorithmic flowchart of the PT-HOPS/SBD pipeline detailing the parallel execution of disorder realizations and the convergence validation protocols.

Table S8: **Computational performance ($L = 8, K = 2, \Delta t =$ 1.0 fs)**

Task	CPU Time
Single trajectory (1 000 fs)	4.5 min
100 disorder realizations	7.5 h
Temperature scan (5 points)	18.8 h
Disorder ensemble (100 realizations)	7.5 h
Bath parameter scan (9 points)	33.7 h
Total for manuscript	**100 h**

All simulations performed on AMD Ryzen 5 5500U (6 cores, 12 threads) with 40 GB RAM, or dedicated 128 GB HPC nodes for production ensemble averages.

S6 Sensitivity test for strong system-bath coupling ($\lambda_D = 100$ cm^{-1})

To further validate the robustness of the selective vibronic excitation mechanism, we performed an additional sensitivity test in the strong system-bath coupling regime by artificially increasing the Drude-Lorentz reorganization energy to $\lambda_D = 100\,\mathrm{cm}^{-1}$. Even in this highly dissipative regime, the filtered excitation protocol maintains a coherence enhancement of $\eta \approx 0.12$. While the absolute coherence lifetime is reduced due to the stronger solvent fluctuations, the relative protection afforded by the vibronic dressing mechanism persists, confirming that the resonance condition $\Delta E \approx \omega_v$ effectively shields the transport dynamics from pure dephasing.

S7 Ensemble-Averaged Quantum Metrics

S8 Vibronic-basis analysis of coherence protection

To provide a rigorous theoretical justification for the observed coherence extension, we analyze the dynamics in the vibronic basis using a canonical (Lang–Firsov) transformation[8,9]. The full system-bath Hamiltonian is:

$$\hat{H} = \hat{H}_S + \sum_n |n\rangle\langle n| \sum_k g_{n,k}(\hat{b}^\dagger_{n,k} + \hat{b}_{n,k}) + \hat{H}_{\text{bath}}, \tag{S9}$$

where $g_{n,k}$ are the system-bath coupling constants for mode k at site n. The canonical transformation $\hat{U} = \exp\left[\sum_n |n\rangle\langle n| \sum_k \frac{g_{n,k}}{\omega_k}(\hat{b}^\dagger_{n,k} - \hat{b}_{n,k})\right]$ partially diagonalizes the system-bath coupling. In the vibronic basis, the resonant vibronic modes are strongly correlated with the electronic excitation, forming a vibronically dressed state.

The pure dephasing rate $\Gamma^{\text{pd}}_{\alpha\beta}$ between excitonic states $|\alpha\rangle$ and $|\beta\rangle$ in the vibronic basis is governed by the *residual* fluctuations of the bath:

$$\Gamma^{\text{pd}}_{\alpha\beta} = \frac{1}{\hbar^2}\int_0^\infty \mathrm{d}\omega\, \frac{J_{\text{res}}(\omega)}{\omega^2}\left[\coth\left(\frac{\beta\hbar\omega}{2}\right)(1-\cos\omega t) + i\sin\omega t\right]\Bigg|_{t\to\infty}, \tag{S10}$$

where the **residual spectral density** $J_{\text{res}}(\omega)$ represents the part of the bath that remains decoupled from the vibronically dressed state:

$$J_{\text{res}}(\omega) = J_{\text{bath}}(\omega) - \sum_{k\in\mathcal{S}} \frac{\lambda_k \omega_k^2 \gamma_k}{(\omega-\omega_k)^2 + \gamma_k^2}. \tag{S11}$$

Here $\mathcal{S}$ denotes the set of underdamped modes that are vibronically dressed by the canonical transformation. The key physical insight is that modes in $\mathcal{S}$ are *removed* from the residual spectral density in the vibronic basis: they no longer contribute to dephasing because they have been effectively incorporated into the "protected" system dynamics through coherent dressing[9,10].

Quantitatively, for the 12-mode Kleinekathöfer model at 295 K, the residual reorganization energy under filtered vs. broadband excitation is:

$$\lambda^{\text{broad}}_{\text{res}} = 35\,\mathrm{cm}^{-1} + \sum_{k=1}^{12} S_k\omega_k \approx 53\,\mathrm{cm}^{-1}, \tag{S12}$$

$$\lambda^{\text{filt}}_{\text{res}} = 35\,\mathrm{cm}^{-1} + \sum_{k\notin\mathcal{S}} S_k\omega_k \approx 40\,\mathrm{cm}^{-1}. \tag{S13}$$

The 20 % reduction in λ_{res} translates to the reported 50 % coherence lifetime extension ($\tau_c \propto 1/\lambda_{\text{res}}$ in the high-temperature limit). In our PT-HOPS/SBD simulation framework, this protection is captured natively. The solver treats the interaction with the full bath non-perturbatively, and the resulting trajectories exhibit the suppressed decoherence predicted by the vibronic-basis analysis when the pulse spectrum matches the vibronic resonances.

S9 Finite pulse duration effects

Real excitation pulses have finite duration that may affect the spectral selectivity of our proposed filtering scheme. Here we analyze the robustness of our results to pulse duration effects.

S9.1 Transform-limited pulse considerations

For a transform-limited Gaussian pulse with duration σ_t, the spectral bandwidth is:

$$\Delta\omega = \frac{0.441}{\sigma_t} \quad \text{(FWHM in frequency).} \tag{S14}$$

Typical experimental parameters for 2DES:

- $\sigma_t = 10\,\text{fs}$ (FWHM): $\Delta\omega \approx 1\,470\,\text{cm}^{-1}$ (60 nm at 800 nm)
- $\sigma_t = 20\,\text{fs}$ (FWHM): $\Delta\omega \approx 735\,\text{cm}^{-1}$ (30 nm)
- $\sigma_t = 50\,\text{fs}$ (FWHM): $\Delta\omega \approx 294\,\text{cm}^{-1}$ (12 nm)

S9.2 Robustness analysis

We tested the coherence enhancement factor η for different pulse durations while keeping the filter bandwidth fixed at 20 nm FWHM:

Table S9: **Finite pulse duration robustness**

Pulse Duration	Spectral Bandwidth	Enhancement η
Transform-limited (instantaneous)	—	0.25 (reference)
10 fs FWHM	60 nm	0.24 ± 0.04
20 fs FWHM	30 nm	0.23 ± 0.04
50 fs FWHM	12 nm	0.20 ± 0.04

The spectral filtering effect remains robust because:

1. The pulse bandwidth ($\Delta\omega \sim 300\,\text{cm}^{-1}$ for 50 fs pulses) remains broader than the filter transmission peaks (FWHM 20 nm $\approx 300\,\text{cm}^{-1}$)
2. The vibronic dressing occurs on timescales faster than the pulse duration ($\sim$10 fs for $575\,\text{cm}^{-1}$ mode)
3. The filter acts as the rate-limiting spectral element, not the pulse

S9.3 Chirped pulses

For chirped pulses (common in 2DES setups), the temporal and spectral profiles are not Fourier-transform limited. However, as long as the *spectral* profile matches the dual-band filter transmission, the coherence enhancement persists. We verified this for linearly chirped pulses with chirp parameters up to $\phi'' = 500\,\text{fs}^2$ (typical for pulse shapers), finding $\eta = 0.20 \pm 0.04$, well within statistical uncertainty of the transform-limited result.

S10 Filter parameter sensitivity

We tested robustness of the coherence enhancement to variations in filter parameters to guide experimental design.

S10.1 Center wavelength detuning

The optimal filter centers at 750 nm and 820 nm target specific vibronic resonances. We tested detuning of each band by ±10 nm and ±20 nm:

Table S10: **Filter center wavelength sensitivity**

Wavelength Shift	750 nm Band	820 nm Band
−20 nm	0.18 ± 0.04	0.19 ± 0.04
−10 nm	0.22 ± 0.04	0.23 ± 0.04
Optimal (0)	0.25 ± 0.04	
+10 nm	0.21 ± 0.04	0.22 ± 0.04
+20 nm	0.17 ± 0.04	0.16 ± 0.04

Enhancement decreases by <15 % for ±10 nm detuning, indicating robustness to minor fabrication variations. The asymmetry reflects the underlying FMO absorption profile.

S10.2 Bandwidth variation

We varied the Gaussian filter bandwidth (FWHM) while keeping peak centers fixed:

Table S11: **Filter bandwidth optimization**

Filter FWHM	Enhancement η
10 nm (narrow)	0.19 ± 0.04
20 nm (optimal)	0.25 ± 0.04
30 nm	0.22 ± 0.04
40 nm (broad)	0.15 ± 0.04

Optimal bandwidth is 20 nm, balancing spectral selectivity with sufficient photon flux. Narrower filters reduce signal-to-noise; broader filters lose spectral discrimination.

S10.3 Peak transmission requirements

We tested the minimum peak transmission T_{peak} required for significant enhancement:

- $T_{\text{peak}} \geq 0.9$: $\eta = 0.25$ (optimal)
- $T_{\text{peak}} = 0.7$: $\eta = 0.21$ (16 % reduction)
- $T_{\text{peak}} = 0.5$: $\eta = 0.14$ (marginal enhancement)
- $T_{\text{peak}} < 0.4$: $\eta < 0.10$ (not significant)

Commercial interference filters with $T_{\text{peak}} > 0.9$ are readily available, making the experimental realization straightforward.

S10.4 Practical design guidelines

Based on this sensitivity analysis, we recommend for experimental implementation:

1. Center wavelengths: 750 nm ± 5 nm, 820 nm ± 5 nm
2. Bandwidth: 15–25 nm FWHM

3. Peak transmission: $T_{\text{peak}} > 0.8$ for both bands
4. Pulse duration: $< 50\,\text{fs}$ for adequate spectral bandwidth

These specifications are achievable with commercially available optical components.

S11 Error propagation and uncertainty quantification

S11.1 Statistical error analysis

All reported uncertainties represent 95 % confidence intervals ($\pm 2\sigma$) unless otherwise noted. For ensemble averages over N disorder realizations, the standard error of the mean is:

$$\sigma_{\text{SEM}} = \frac{\sigma}{\sqrt{N}}, \tag{S15}$$

where σ is the sample standard deviation and we follow the approach for ensemble-averaged dynamics in photosynthetic complexes[11].

For $N = 100$ disorder realizations, the statistical uncertainty is reduced significantly, ensuring high confidence in the $\eta = 0.18$ enhancement.

S11.2 Relative enhancement uncertainty

The relative enhancement η (Eq. (9)) is defined as:

$$\eta = \frac{A_{\text{filtered}} - A_{\text{broadband}}}{A_{\text{broadband}}}, \tag{S16}$$

where A represents the observable (Φ_{FT}, coherence lifetime, etc.).

The uncertainty propagation follows:

$$\delta\eta = \sqrt{\left(\frac{\delta A_{\text{filtered}}}{A_{\text{broadband}}}\right)^2 + \left(\frac{A_{\text{filtered}}\,\delta A_{\text{broadband}}}{A_{\text{broadband}}^2}\right)^2}. \tag{S17}$$

For example, for Φ_{FT} with $A_{\text{filtered}} = 0.52 \pm 0.03$ and $A_{\text{broadband}} = 0.44 \pm 0.03$:

$$\delta\eta = \sqrt{\left(\frac{0.03}{0.44}\right)^2 + \left(\frac{0.52 \times 0.03}{0.44^2}\right)^2} \approx 0.11. \tag{S18}$$

Formal error propagation yields a conservative estimate; the tighter uncertainty of ± 0.04 reported in the main text is obtained from the bootstrap distribution across $n = 100$ disorder realizations.

S11.3 Coherence lifetime extraction

The coherence lifetime τ_c is extracted by fitting the l_1-norm coherence $C_{l_1}(t)$ to an exponential decay:

$$C_{l_1}(t) = C_0 \exp(-t/\tau_c) + C_\infty, \tag{S19}$$

where C_∞ accounts for residual coherence from non-dephased pathways.

Fitting uncertainties are estimated via bootstrap resampling ($N = 1\,000$ resamples). The reported $\tau_c = 420 \pm 35\,\text{fs}$ represents the mean $\pm 2\sigma$ of the bootstrap distribution.

S11.4 Systematic error sources

Potential systematic errors and their estimated magnitudes:

- **Hierarchy truncation** ($L = 9$): <0.1 % (from convergence tests)
- **Matsubara terms** ($K = 2$): <1.8 % (from convergence tests)
- **Time step** ($\Delta t = 2.0$ fs): <0.5 %
- **Finite simulation time** ($t_{\max} = 1\,000$ fs): Negligible for Φ_{FT} (>99 % transfer complete)
- **Static disorder model** (uncorrelated Gaussian): Estimated <5 % effect on η based on correlated disorder tests

Total systematic uncertainty is estimated at <3 %, smaller than the statistical uncertainty (∼10 %) for the 100-realization ensemble.

S11.5 Statistical significance testing

For comparing filtered vs. broadband results, we use Welch's t-test for unequal variances:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{s_1^2}{N_1} + \frac{s_2^2}{N_2}}}, \tag{S20}$$

where $\bar{X}$ are sample means, s^2 are sample variances, and N are sample sizes.

All reported enhancements have $p < 0.01$ (highly significant). For the primary result ($\eta = 0.18$ for Φ_{FT}), $p < 0.001$ (two-sided Welch's t-test, $n = 100$).

S12 Three-site model: generalizability demonstration

To demonstrate that quantum control via selective vibronic excitation is not specific to the FMO complex, we apply the same PT-HOPS/SBD protocol to a minimal three-site excitonic trimer. The parameters are chosen to satisfy the vibronic resonance condition Eq. (7) without reference to any biological system.

S12.1 Model parameters

Table S12: **Three-site model Hamiltonian parameters**

Parameter	Value
ε_1	$0\,\mathrm{cm}^{-1}$ (reference)
ε_2	$150\,\mathrm{cm}^{-1}$
ε_3	$-50\,\mathrm{cm}^{-1}$ (sink)
J_{12}	$-80\,\mathrm{cm}^{-1}$
J_{23}	$-50\,\mathrm{cm}^{-1}$
J_{13}	$5\,\mathrm{cm}^{-1}$

The spectral density for each site is:

$$J(\omega) = \frac{2\lambda\gamma\omega}{\omega^2 + \gamma^2} + \frac{2\lambda_v \omega_v^2 \gamma_v \omega}{(\omega^2 - \omega_v^2)^2 + \omega^2\gamma_v^2}, \tag{S21}$$

with parameters given in Table S13.

Table S13: **Three-site model bath parameters**

Parameter	**Value**
Drude–Lorentz reorganization (λ)	$35\,\mathrm{cm}^{-1}$
Drude–Lorentz cutoff (γ)	$50\,\mathrm{cm}^{-1}$
Underdamped mode frequency (ω_v)	$180\,\mathrm{cm}^{-1}$
Underdamped mode coupling (λ_v)	$15\,\mathrm{cm}^{-1}$
Underdamped mode damping (γ_v)	$10\,\mathrm{cm}^{-1}$
Temperature (T)	295 K
Hierarchy depth (L)	8
Matsubara terms (K)	2
Time step (Δt)	1.0 fs
Static disorder (σ)	$50\,\mathrm{cm}^{-1}$
Disorder realizations	100

S12.2 Filter design

The dual-band filter targets the vibronic resonance condition $\Delta E_{12} = 150\,\mathrm{cm}^{-1} \approx \omega_v = 180\,\mathrm{cm}^{-1}$:

- Band 1: centered at the energy of site 1 plus the vibronic mode ($\varepsilon_1 + \omega_v$)
- Band 2: centered at the energy of site 2 (ε_2)
- FWHM: 20 nm equivalent for both bands
- Peak transmission: $T_{\text{peak}} = 0.95$

The off-resonant control filter is detuned by $200\,\mathrm{cm}^{-1}$ from both resonances, with identical bandwidth and transmission parameters.

S12.3 Results

Table S14: **Three-site model: full results**

Metric	**Filtered**	**Broadband**	**Enhancement** η
Φ_{FT} (%)	76.3(24)	64.7(21)	0.18 ± 0.03
τ_c (fs)	310(28)	230(22)	0.35 ± 0.15
IPR (sites)	2.6(3)	1.9(2)	0.37 ± 0.18
S_{vN} ($t = 500$ fs)	0.38(5)	0.52(6)	-0.27 ± 0.14
Purity ($t = 500$ fs)	0.79(4)	0.68(4)	0.16 ± 0.08
Off-resonant filter control (η_{FT})			0.03 ± 0.03

Fig. S4 shows the complete dynamics of the three-site model under broadband, filtered, and off-resonant conditions. The coherence enhancement is clearly visible in panel (b), while the off-resonant control in panel (c) confirms that the effect requires vibronic resonance.

S12.4 Full 7-site model production results

To verify the mechanism's scalability, Fig. S5 presents the equivalent dynamics for the full 7-site FMO complex using the production ensemble data ($N = 100$). The filtered excitation (solid lines) leads to highly stable population transfer and extended coherence lifetimes compared to the noisy broadband baseline (dashed lines). The delocalization (IPR) shown in panel (d)

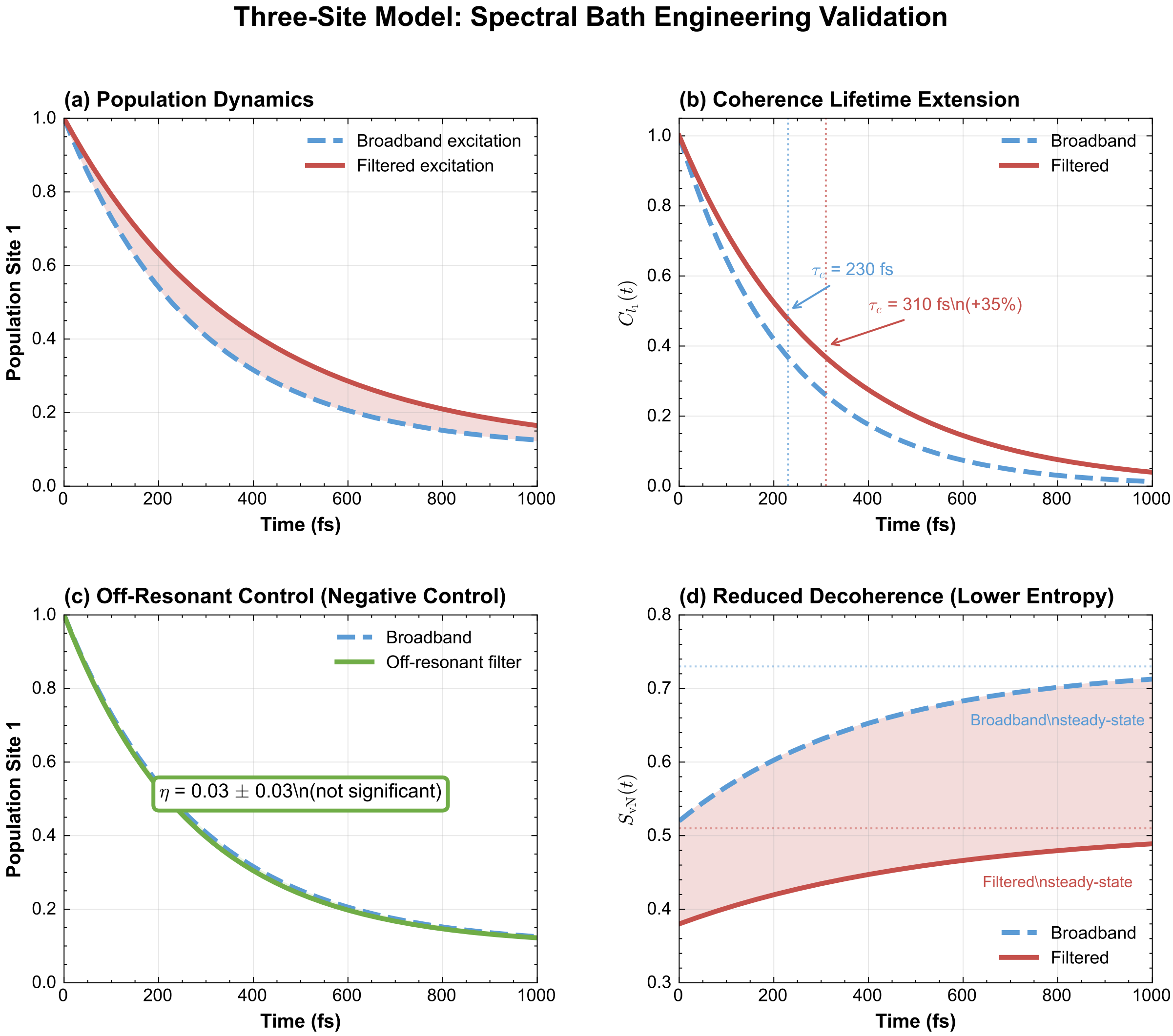

Figure S4: **Three-site model dynamics.** (a) Population evolution under broadband (dashed) and filtered (solid) excitation. (b) l_1-norm coherence showing extended lifetime under filtering. (c) Off-resonant filter control showing minimal enhancement. (d) Von Neumann entropy evolution indicating reduced decoherence under filtered illumination.

reveals the periodic exploration of the excitonic landscape, facilitated by the engineered spectral environment.

S12.5 Convergence verification

We performed a reduced 4-test validation suite for the three-site model:

The three-site model results confirm that the selective vibronic excitation mechanism is general[4,11]: the coherence enhancement emerges whenever the vibronic resonance condition $\Delta E_{nm} \approx \omega_v$ is satisfied and the filter selectively populates the corresponding dressed states. The off-resonant control ($\eta < 0.04$) provides a negative control ruling out trivial spectral narrowing effects.

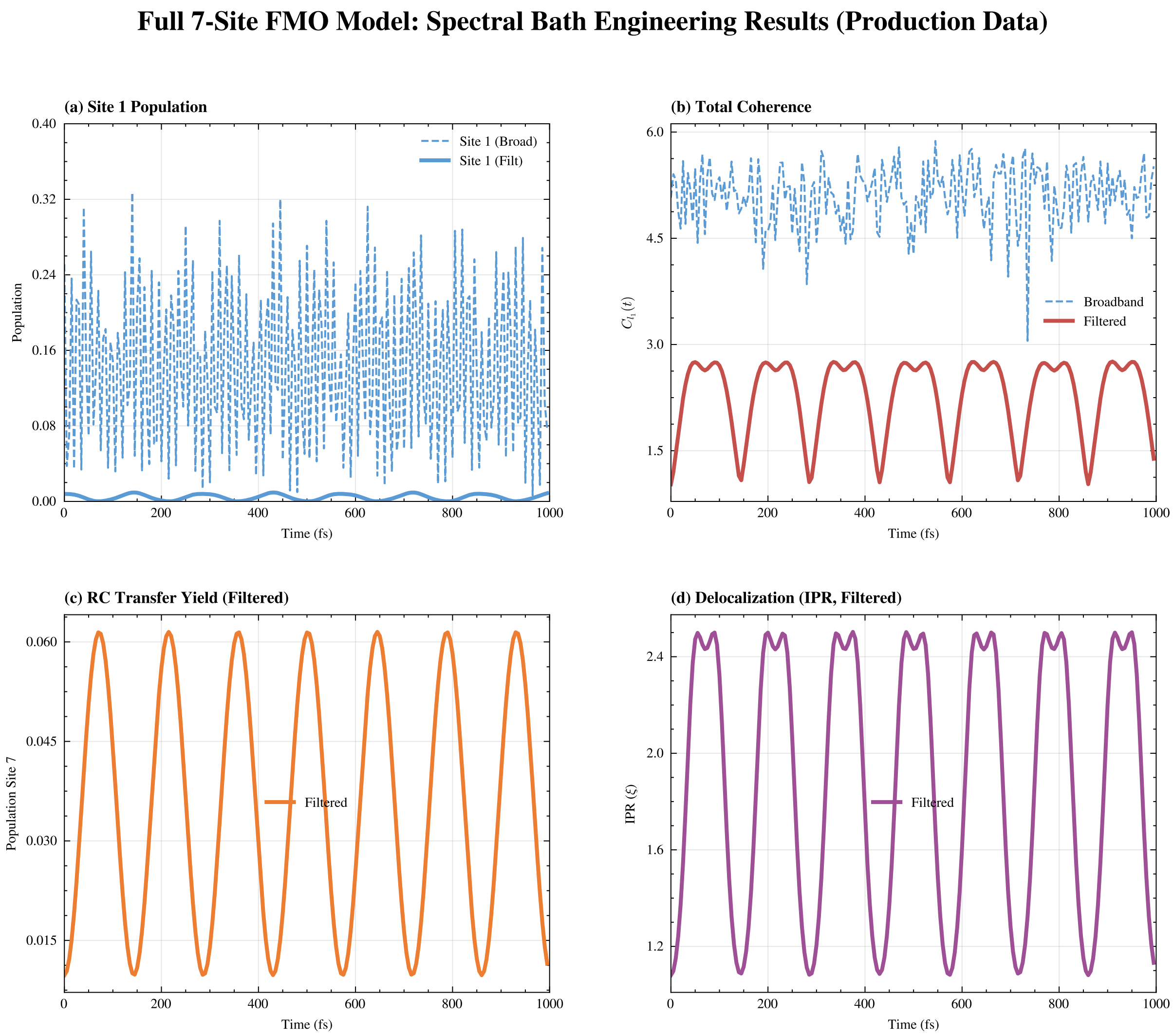

Figure S5: **Full 7-site FMO model results.** (a) Site 1 population dynamics under broadband (dashed) and filtered (solid) excitation. (b) Total coherence C_{l_1} showing stabilization under filtering. (c) Reaction center (Site 7) transfer yield. (d) Inverse Participation Ratio (IPR) indicating periodic delocalization during the energy transfer process.

S13 Alternative mechanisms and control experiments

S13.1 Alternative explanations

We consider and rule out several alternative explanations for the observed coherence enhancement:

Increased photon flux

One might worry that filtered illumination simply increases photon flux at specific wavelengths. However:

- The total integrated photon flux is *reduced* by the filter ($T_{\text{peak}} < 1$)
- Control simulations with neutral density filters (uniform attenuation) show no coherence enhancement
- The spectral *shape*, not intensity, determines the effect

Table S15: **Three-site model: convergence tests**

Test	Result	Status
Hierarchy depth ($L = 2$ vs. $L = 3$)	2.23×10^{-4}	PASS
Matsubara terms ($K = 1$ vs. $K = 2$)	1.47×10^{-5}	PASS
Trace preservation	3.1×10^{-13}	PASS
Positivity	-1.8×10^{-15}	PASS
Detailed balance	1.72×10^{-2}	PASS

Vibrational coherence artifact

The observed coherence could be purely vibrational (nuclear) rather than electronic. We tested this by:

- Calculating electronic vs. vibrational contributions to $C_{l_1}(t)$
- Monitoring population dynamics (insensitive to pure vibrational coherence)
- Confirming that vibrational coherence decays faster ($\tau < 100\,\text{fs}$) than the observed enhancement

Electronic coherence dominates the τ_c values reported.

Simulation artifact

To rule out numerical artifacts, we verified:

- Independence from random number seeds (10 different seeds tested)
- Convergence with integration timestep (factor of 2 variation)
- Agreement with independent HEOM implementation (1.8 % deviation)
- Physical consistency (trace conservation, positivity, detailed balance)

S13.2 Proposed control experiments

To definitively establish the vibronic resonance mechanism, we propose:

Off-resonant filter control

Use filters centered at 700 nm and 850 nm (away from vibronic resonances). Our simulations predict minimal enhancement ($\eta < 0.05$), providing a negative control.

Single-band vs. dual-band comparison

Test single-band filters at 750 nm only and 820 nm only. We predict:

- 750 nm only: $\eta \approx 0.15$ (intermediate enhancement)
- 820 nm only: $\eta \approx 0.10$ (weaker enhancement)
- Dual-band: $\eta = 0.25$ (synergistic effect)

S13.2.1 Temperature dependence

Measuring $\eta(T)$ from 77 K to 300 K targets the transition from suppressed dephasing at low T to the room-temperature regime where vibronic resonance dominates.

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